PROJET SISE-OPSIE 2026

LAKHAL, JABIL, PECNIK, KNOEPFFLER

Présentation

Données

Analyses

Résultats

Présentation du projet

Fouille de données et détection des intrusions.

Contenu du projet :

- Analyses des données fournies de type IPTABLES
- Web App Python
- Machine Learning

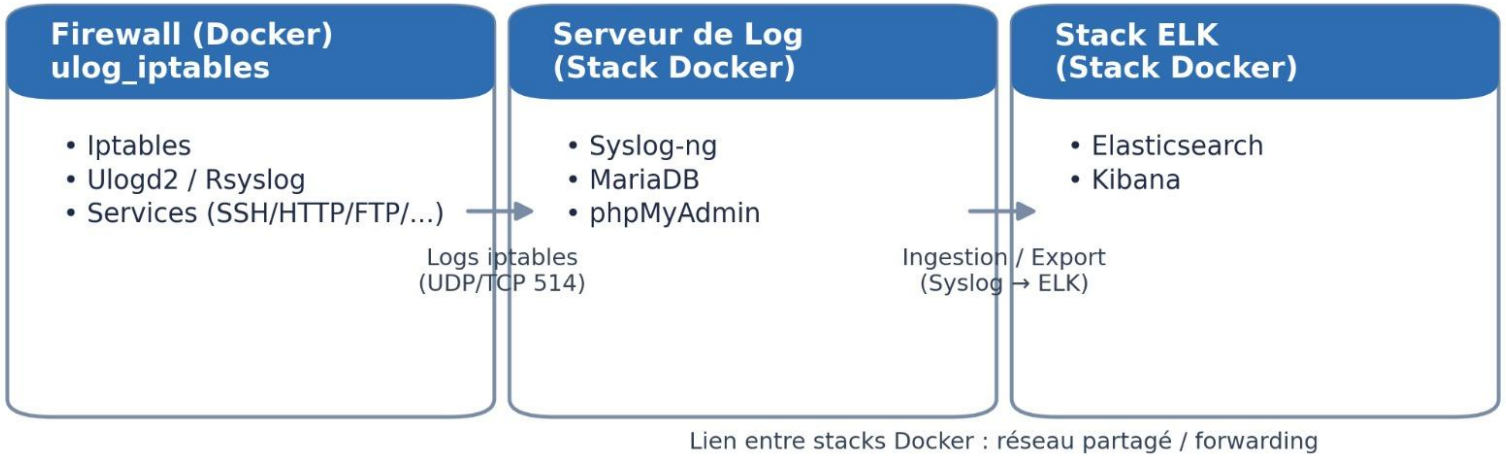
Présentation des données

timestamp		hostname		action		# rule		interface_in	
Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)
Distinct:	3106616 (68%)	Distinct:	1 (<1%)	Distinct:	2 (<1%)	Distinct:	10 (<1%)	Distinct:	1 (<1%)
Min:	2025-11-12 09:56:37	159.84.146.99	100%	PERMIT	82%			eth0	100%
Max:	2026-02-12 10:08:39			DENY	18%				

src_ip		dst_ip		# len		# ttl		# id		df	
Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)
Distinct:	130849 (3%)	Distinct:	1 (<1%)	Distinct:	181 (<1%)	Distinct:	248 (<1%)	Distinct:	65536 (1%)	False:	602443 (13%)
3.224.220.101	17%	159.84.146.99	100%							True:	3970460 (87%)
23.22.35.162	17%										
52.70.240.171	15%										
Other	51%										

proto		# src_port		# dst_port		# window		flags	
Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)	Missing:	0 (0%)
Distinct:	2 (<1%)	Distinct:	64566 (1%)	Distinct:	65532 (1%)	Distinct:	12285 (<1%)	Distinct:	9 (<1%)
TCP	100%							SYN	98%
	0%							CWR,ECE,SYN	1%
								ACK	<1%
								Other	<1%

Architecture / Environment



Connexions licites

```
root@ddd375a4f681:/# chmod +x trafic_ok.sh
root@ddd375a4f681:/# ./trafic_ok.sh
[2026-03-02 18:32:57] ===== Démarrage trafic licite (45 min) - cible 172.17.0.3  =====
[2026-03-02 18:32:57] SCAN TCP 1-65535 début...
[2026-03-02 18:33:00] SCAN TCP 1-65535 terminé
[2026-03-02 18:33:00] --- Cycle 1 (restant : 2697s) ---
[2026-03-02 18:33:00] HTTP /Restore FAIL
[2026-03-02 18:33:04] PORT 3389 (RDP) OK
[2026-03-02 18:33:06] HTTP /Backup FAIL
[2026-03-02 18:33:10] HTTP / FAIL
[2026-03-02 18:33:14] PORT 443 (HTTPS) OK
[2026-03-02 18:33:18] PORT 8080 (HTTP-alt) OK
[2026-03-02 18:33:19] PORT 110 (POP3) OK
[2026-03-02 18:33:20] PORT 445 (SMB) OK
[2026-03-02 18:33:22] SSH port knock OK
[2026-03-02 18:33:23] TELNET OK
[2026-03-02 18:33:25] FTP anonymous OK
[2026-03-02 18:33:27] SSH port knock OK
[2026-03-02 18:33:31] SSH port knock OK
[2026-03-02 18:33:33] SSH port knock OK
[2026-03-02 18:33:54] --- Cycle 2 (restant : 2643s) ---
[2026-03-02 18:33:54] PORT 8080 (HTTP-alt) OK
[2026-03-02 18:33:58] PORT 110 (POP3) OK
[2026-03-02 18:33:59] PORT 445 (SMB) OK
[2026-03-02 18:34:02] FTP anonymous OK
```

Connexions licites

```
root@ddd375a4f681:/# chmod +x trafic_ok.sh
root@ddd375a4f681:/# ./trafic_ok.sh
[2026-03-02 18:32:57] ===== Démarrage trafic licite (45 min) - cible 172.17.0.3  =====
[2026-03-02 18:32:57] SCAN TCP 1-65535 début...
[2026-03-02 18:33:00] SCAN TCP 1-65535 terminé
[2026-03-02 18:33:00] --- Cycle 1 (restant : 2697s) ---
[2026-03-02 18:33:00] HTTP /Restore FAIL
[2026-03-02 18:33:04] PORT 3389 (RDP) OK
[2026-03-02 18:33:06] HTTP /Backup FAIL
[2026-03-02 18:33:10] HTTP / FAIL
[2026-03-02 18:33:14] PORT 443 (HTTPS) OK
[2026-03-02 18:33:18] PORT 8080 (HTTP-alt) OK
[2026-03-02 18:33:19] PORT 110 (POP3) OK
[2026-03-02 18:33:20] PORT 445 (SMB) OK
[2026-03-02 18:33:22] SSH port knock OK
[2026-03-02 18:33:23] TELNET OK
[2026-03-02 18:33:25] FTP anonymous OK
[2026-03-02 18:33:27] SSH port knock OK
[2026-03-02 18:33:31] SSH port knock OK
[2026-03-02 18:33:33] SSH port knock OK
[2026-03-02 18:33:54] --- Cycle 2 (restant : 2643s) ---
[2026-03-02 18:33:54] PORT 8080 (HTTP-alt) OK
[2026-03-02 18:33:58] PORT 110 (POP3) OK
[2026-03-02 18:33:59] PORT 445 (SMB) OK
[2026-03-02 18:34:02] FTP anonymous OK
```

Attaques Nmap réalisées

```
Nmap done: 1 IP address (1 host up) scanned in 0.37 seconds
root@ddd375a4f681:/# nmap -sS -Pn -iR 10 --max-hostgroup 10 --max-parallelism 1 172.17.0.3
Starting Nmap 7.80 ( https://nmap.org ) at 2026-03-03 09:09 UTC
Stats: 0:02:58 elapsed; 0 hosts completed (10 up), 10 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.83% done
Stats: 0:04:16 elapsed; 0 hosts completed (10 up), 10 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.21% done; ETC: 14:43 (5:29:17 remaining)
Stats: 0:04:51 elapsed; 0 hosts completed (10 up), 10 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.39% done; ETC: 14:43 (5:28:42 remaining)

root@ddd375a4f681:/# nmap -sS -Pn 172.17.0.3
Starting Nmap 7.80 ( https://nmap.org ) at 2026-03-03 09:16 UTC
Nmap scan report for 172.17.0.3
Host is up (0.000022s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: FA:B2:6E:EA:FA:F3 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.52 seconds
root@ddd375a4f681:/# nmap -sS -T4 -p 1-1000 172.17.0.3
Starting Nmap 7.80 ( https://nmap.org ) at 2026-03-03 09:16 UTC
Nmap scan report for 172.17.0.3
Host is up (0.000037s latency).
```

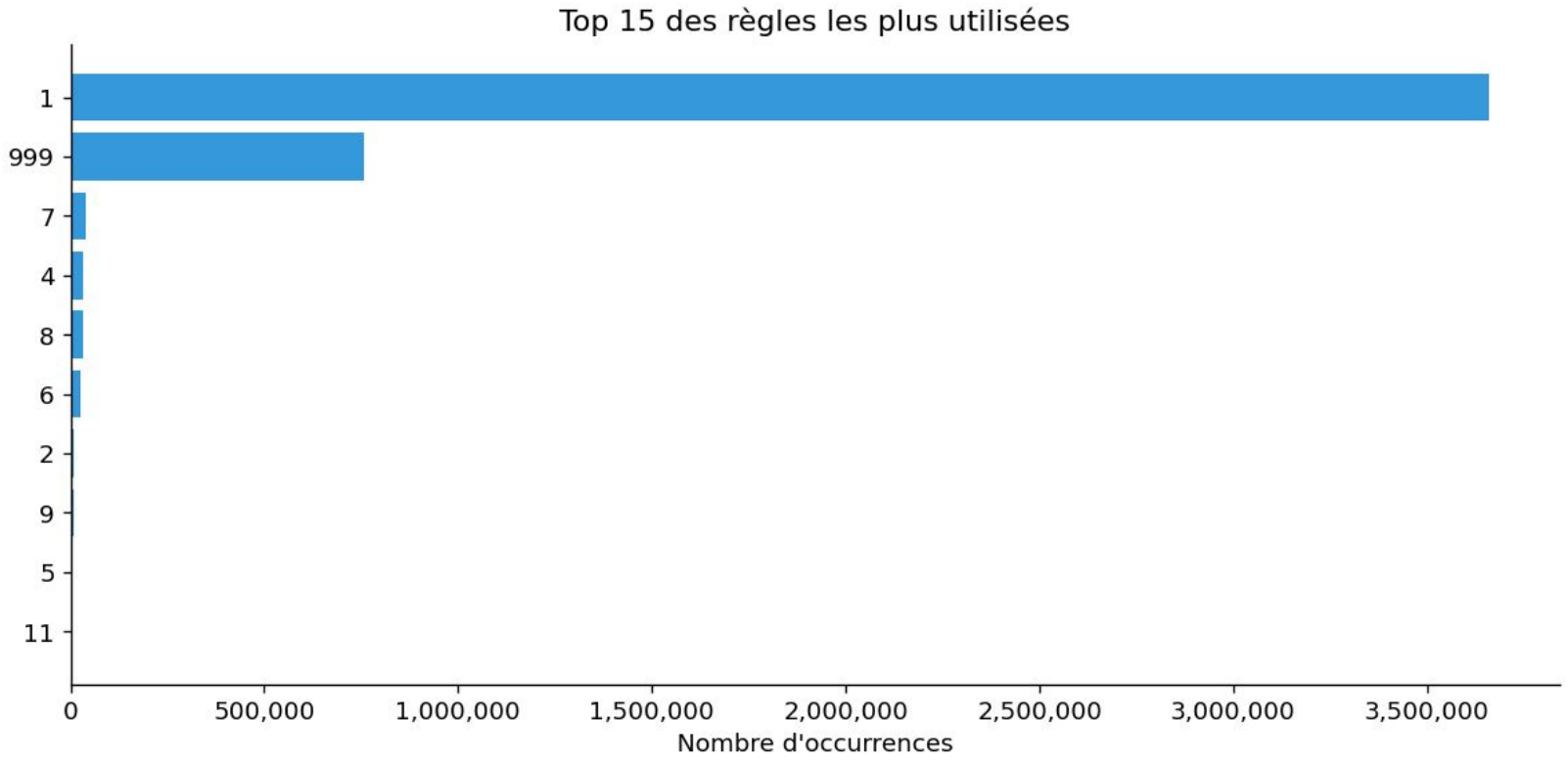
Attaques Nmap réalisées

```
Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds
root@ddd375a4f681:/# nmap -sS -D 10.0.0.1,192.168.4.1,172.156.2.26,192.168.1.0,10.10.1.101
,89.89.56.2,28.12.15.20,10.172.11.2,194.25.56.2,172.5.2.8 -Pn -e eth0 -p 22,23,80,443,3306
  172.17.0.3
Starting Nmap 7.80 ( https://nmap.org ) at 2026-03-03 09:04 UTC
Nmap scan report for 172.17.0.3
```

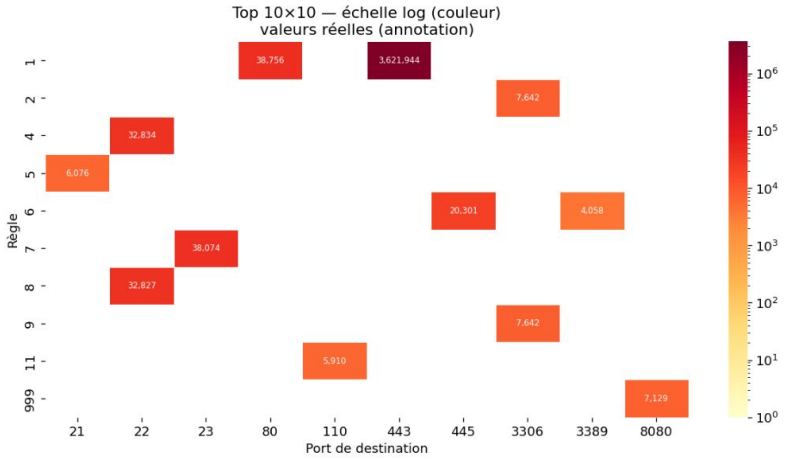
Analyses réalisées

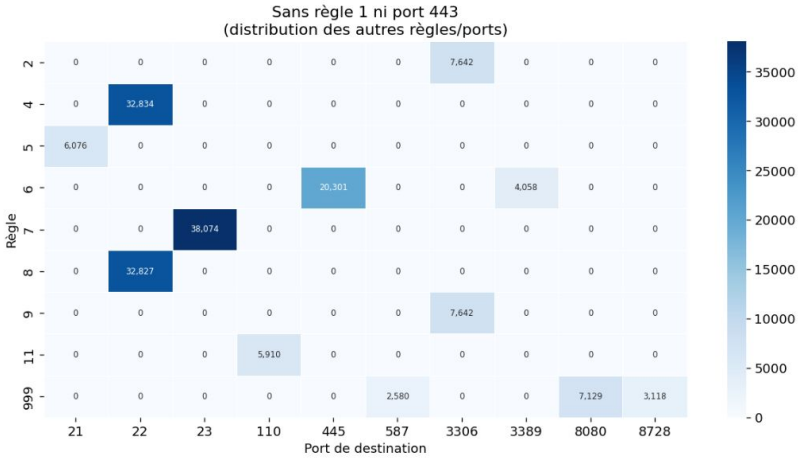
- Notebook
- Machine Learning

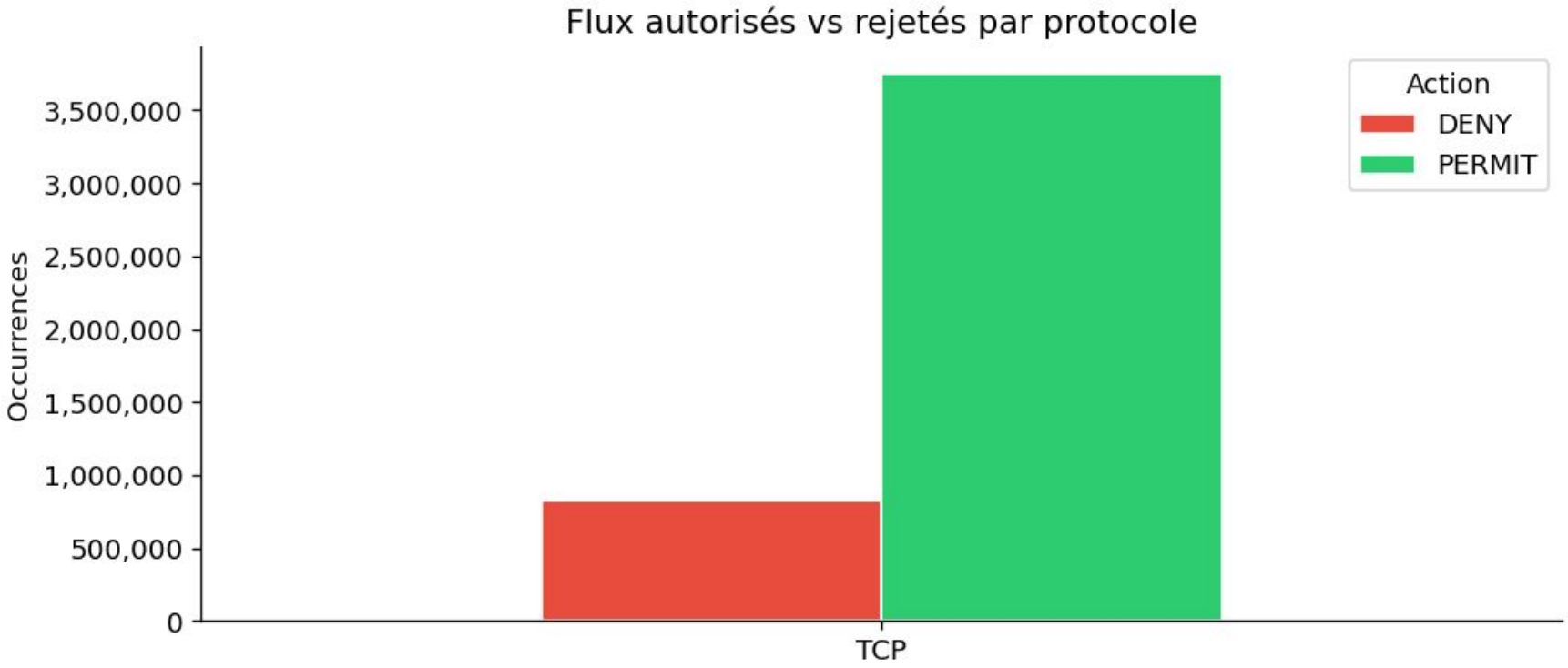
Notebook

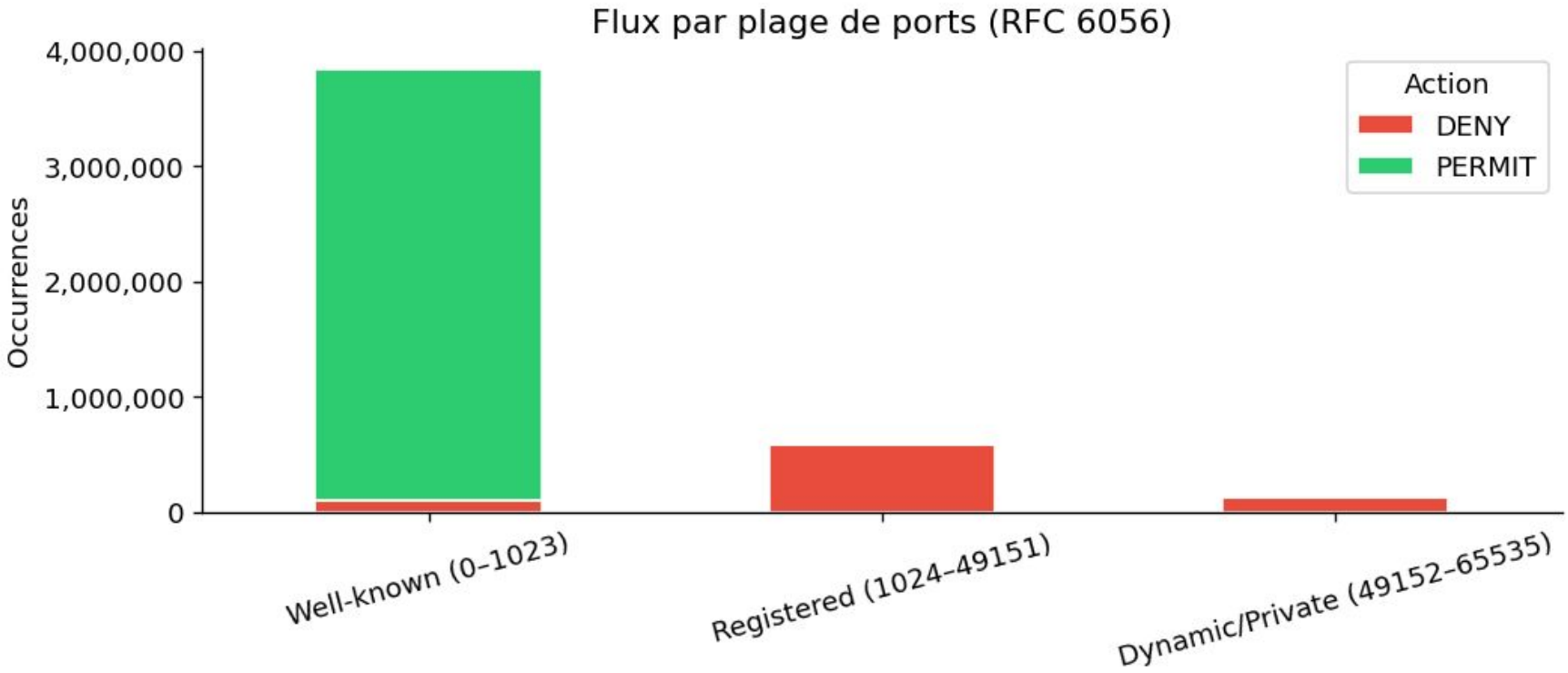

Top 15 des règles les plus utilisées
1
999
7
4
8
6
2
9
5
11
0
500,000
1,000,000
1,500,000
2,000,000
2,500,000
3,000,000
3,500,000
Nombre d'occurrences

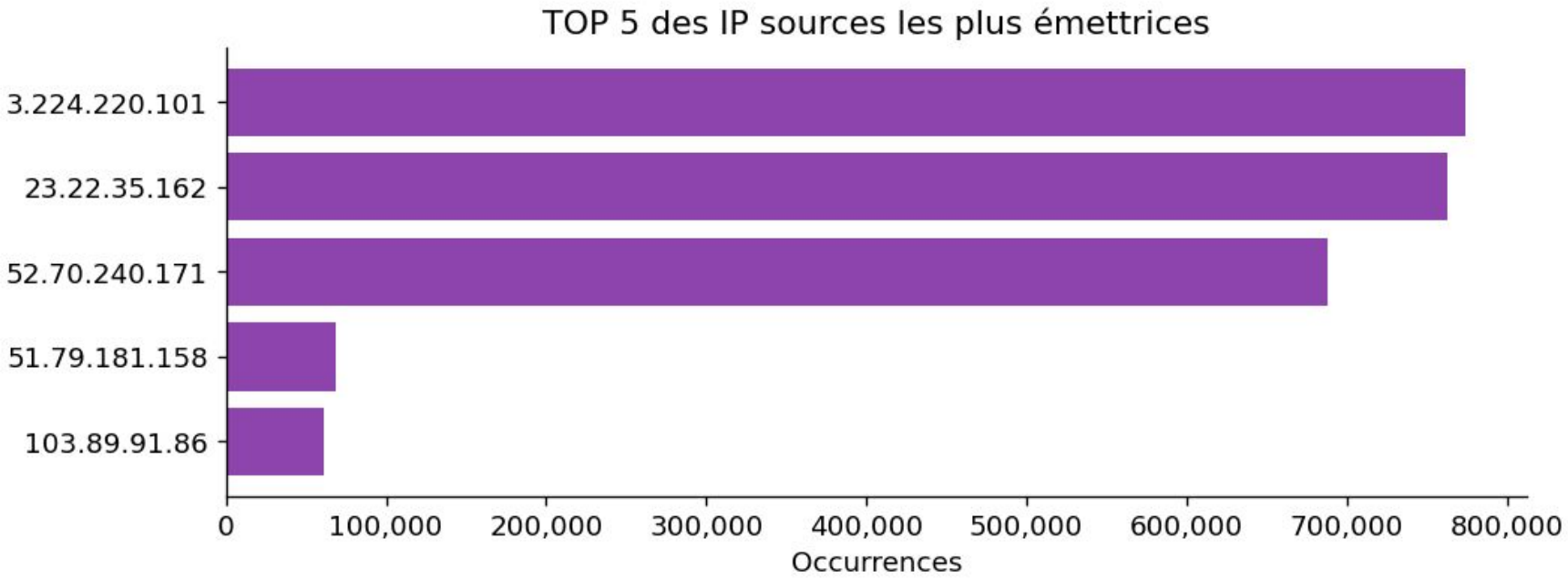
Heatmap règle × port de destination (TCP)

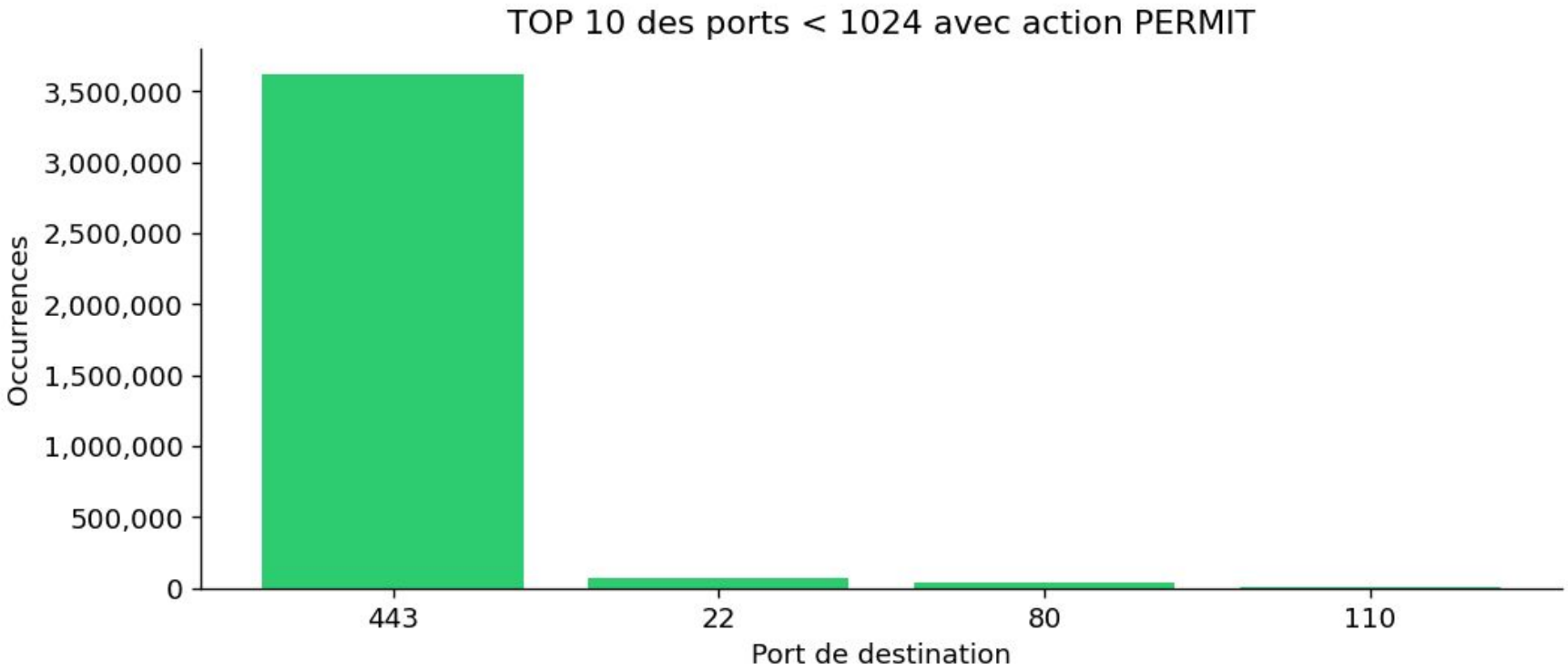


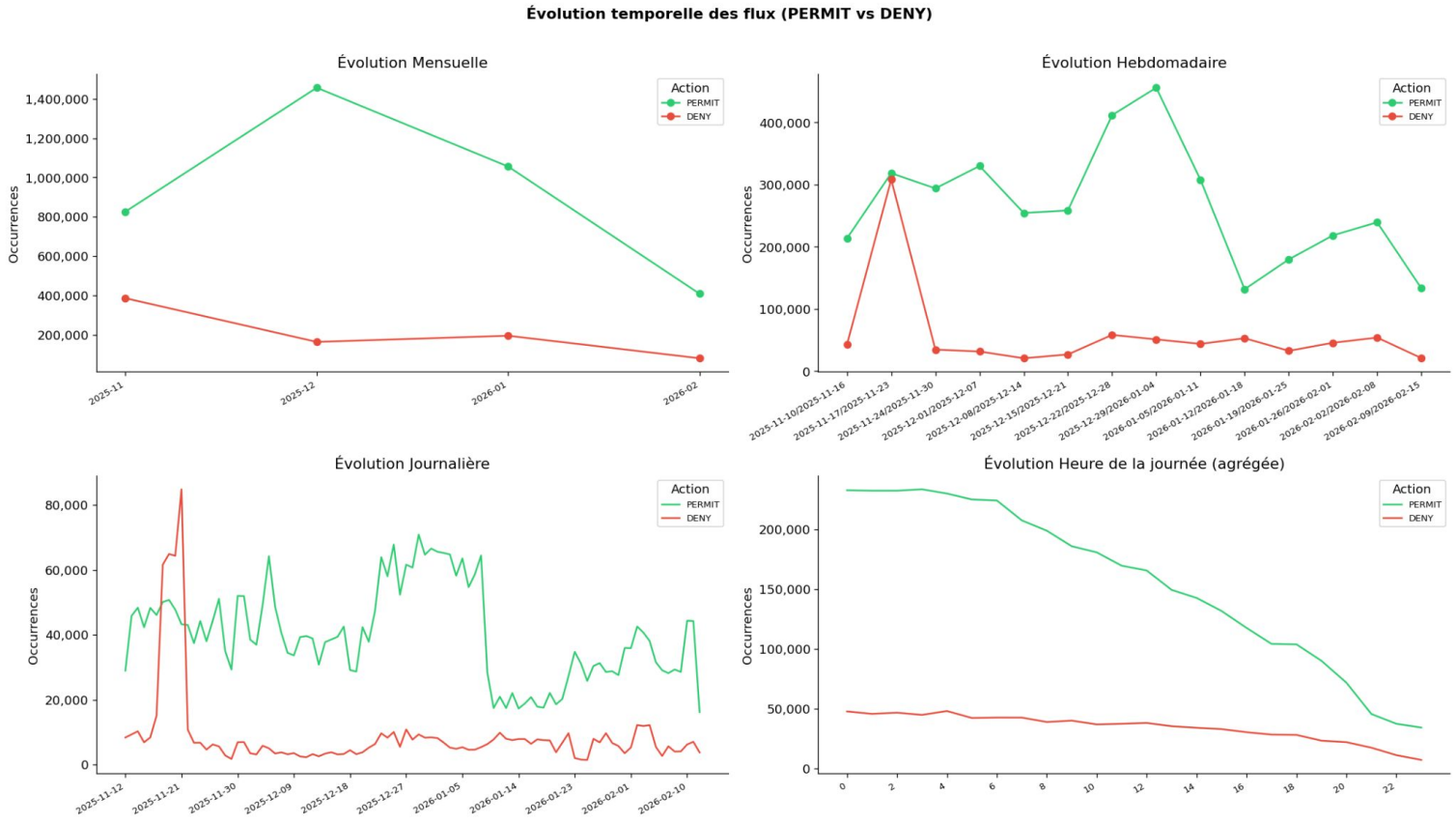


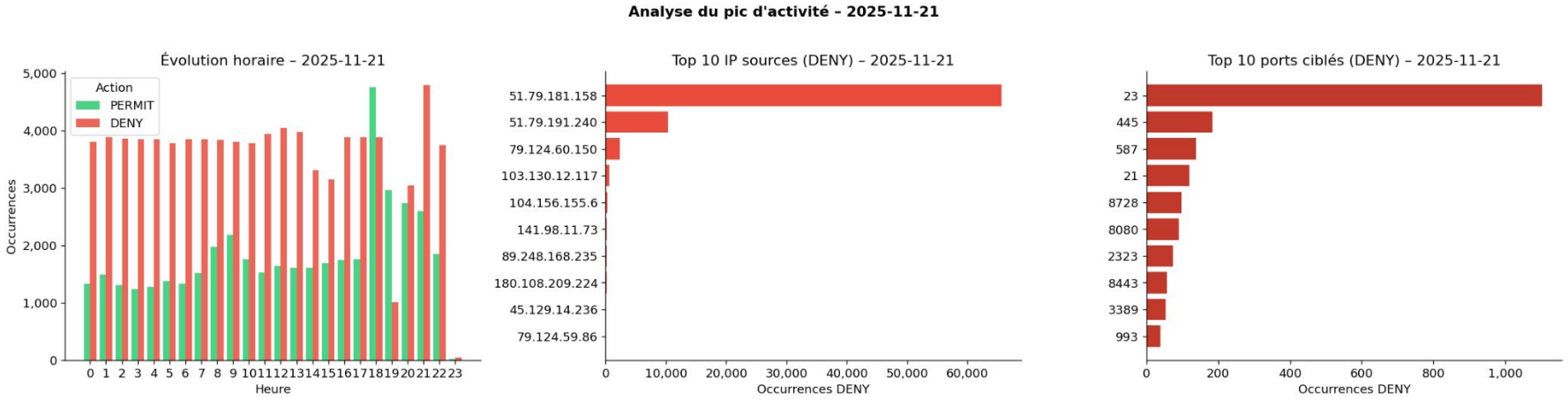

Flux autorisés vs rejetés par protocole
Occurrences
3,500,000
3,000,000
2,500,000
2,000,000
1,500,000
1,000,000
500,000
0
TCP
Action
DENY
PERMIT


Flux par plage de ports (RFC 6056)
Occurrences
4,000,000
3,000,000
2,000,000
1,000,000
0
Action
DENY
PERMIT
Well-known (0–1023)
Registered (1024–49151)
Dynamic/Private (49152–65535)


TOP 5 des IP sources les plus émettrices
3.224.220.101
23.22.35.162
52.70.240.171
51.79.181.158
103.89.91.86
0
100,000
200,000
300,000
400,000
500,000
600,000
700,000
800,000
Occurrences


TOP 10 des ports < 1024 avec action PERMIT
Occurrences
3,500,000
3,000,000
2,500,000
2,000,000
1,500,000
1,000,000
500,000
0
443
22
80
110
Port de destination


Évolution temporelle des flux (PERMIT vs DENY)
Évolution Mensuelle
Occurrences
1,400,000
1,200,000
1,000,000
800,000
600,000
400,000
200,000
2025-11
2025-12
2026-01
2026-02
Action
PERMIT
DENY
Évolution Hebdomadaire
Occurrences
400,000
300,000
200,000
100,000
0
2025-11-10/2025-11-16
2025-11-17/2025-11-23
2025-11-24/2025-11-30
2025-12-01/2025-12-07
2025-12-08/2025-12-14
2025-12-15/2025-12-21
2025-12-22/2025-12-28
2025-12-29/2026-01-04
2026-01-05/2026-01-11
2026-01-12/2026-01-18
2026-01-19/2026-01-25
2026-01-26/2026-02-01
2026-02-02/2026-02-08
2026-02-09/2026-02-15
Action
PERMIT
DENY
Évolution Journalière
Occurrences
80,000
60,000
40,000
20,000
0
2025-11-12
2025-11-21
2025-11-30
2025-12-09
2025-12-18
2025-12-27
2026-01-05
2026-01-14
2026-01-23
2026-02-01
2026-02-10
Action
PERMIT
DENY
Évolution Heure de la journée (agrégée)
Occurrences
200,000
150,000
100,000
50,000
0
0
2
4
6
8
10
12
14
16
18
20
22
Action
PERMIT
DENY

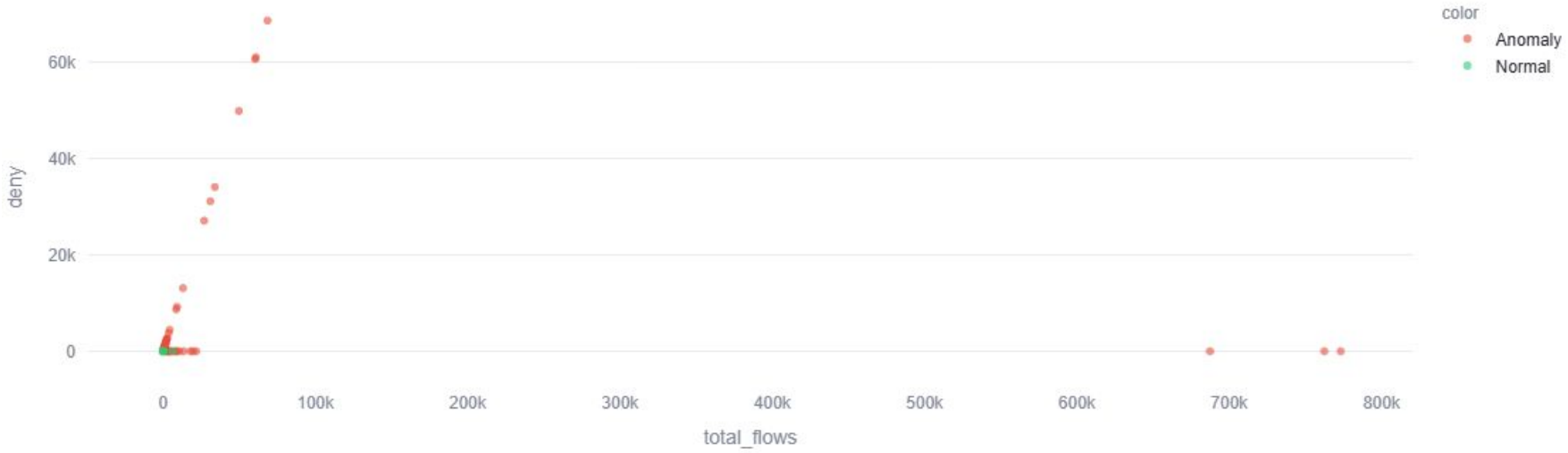

Analyse du pic d'activité – 2025-11-21
Évolution horaire – 2025-11-21
Action
PERMIT
DENY
Occurrences
Heure
Top 10 IP sources (DENY) – 2025-11-21
51.79.181.158
51.79.191.240
79.124.60.150
103.130.12.117
104.156.155.6
141.98.11.73
89.248.168.235
180.108.209.224
45.129.14.236
79.124.59.86
Occurrences DENY
Top 10 ports ciblés (DENY) – 2025-11-21
23
445
587
21
8728
8080
2323
8443
3389
993
Occurrences DENY

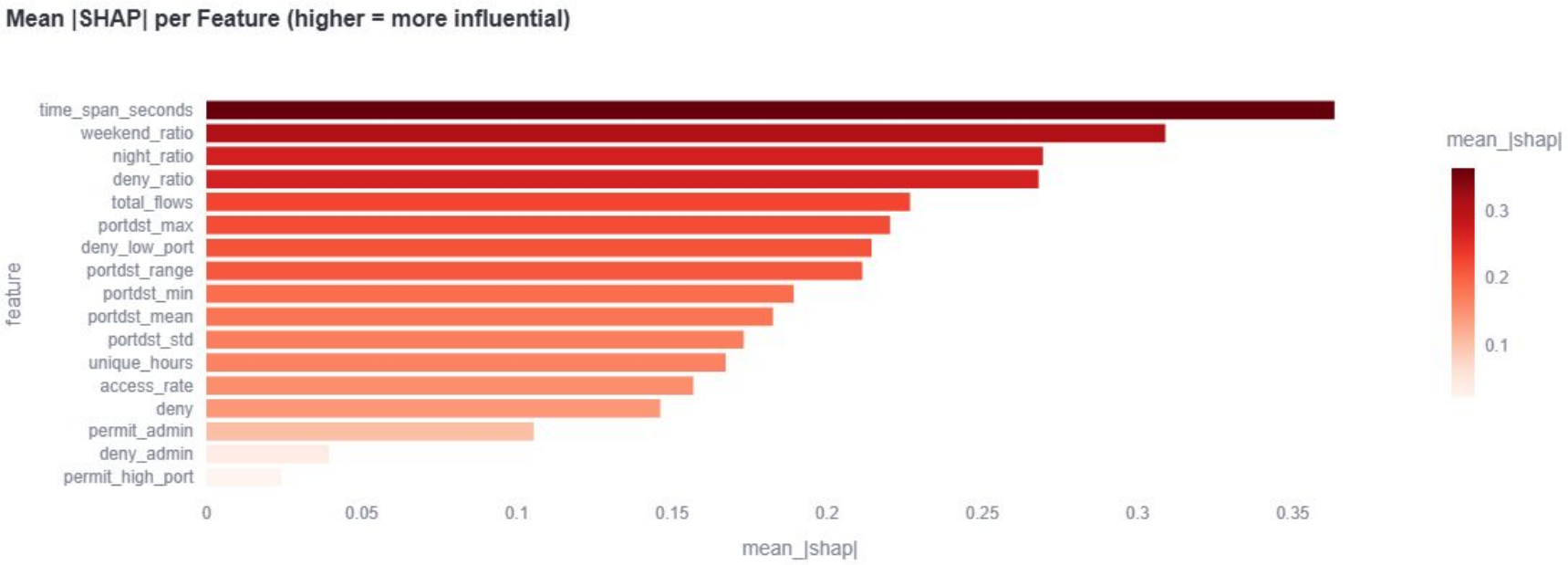
Machine Learning

Total IPs	Anomalies	Normal	Anomaly Rate
130,849	13,085	117,764	10.0%

Anomaly Score Distribution

Anomalies — total_flows vs deny




Mean |SHAP| per Feature (higher = more influential)
time_span_seconds
weekend_ratio
night_ratio
deny_ratio
total_flows
portdst_max
deny_low_port
portdst_range
portdst_min
portdst_mean
portdst_std
unique_hours
access_rate
deny
permit_admin
deny_admin
permit_high_port
feature
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
mean_|shap|
mean_|shap|
0.3
0.2
0.1

Conclusion

✋

Des questions ?